

Supplementary Material:

MacroIRT — Interval-Averaged Bayesian Inference

with State-Dependent Open-Channel Noise

(rank-2 Laplace, two-delta formulation)

1 Setup and notation

We consider N_{ch} independent Markov ion channels with K microscopic states. At the start of an interval $[0, t]$ the macroscopic state is $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$: row-vector mean $\boldsymbol{\mu}_0 \in \mathbb{R}^{1 \times K}$ and per-channel covariance $\boldsymbol{\Sigma}_0 \in \mathbb{R}^{K \times K}$ (singular as ambient $K \times K$, full-rank on the $(K-1)$ -simplex tangent). Microscopic dynamics: rate matrix $\mathbf{Q}$, transition matrix $\mathbf{P}(t) = \exp(\mathbf{Q}t)$.

Per boundary pair (i_0, i_t) , define

$$\bar{\Gamma}(i_0, i_t) = \mathbb{E}[\bar{I}_{0 \rightarrow t} \mid X(0) = i_0, X(t) = i_t], \quad (\text{boundary mean current; } \texttt{gmean_ij}) \quad (1)$$

$$\bar{V}(i_0, i_t) = \text{Var}[\bar{I}_{0 \rightarrow t} \mid X(0) = i_0, X(t) = i_t], \quad (\text{boundary residual variance; } \texttt{gvar_ij}) \quad (2)$$

and their per- i_0 marginals

$$\bar{\gamma}_0(i_0) = \sum_{i_t} P_{i_0 \rightarrow i_t}(t) \bar{\Gamma}(i_0, i_t), \quad (\texttt{gmean_i}) \quad (3)$$

$$\bar{\sigma}^2_0(i_0) = \sum_{i_t} P_{i_0 \rightarrow i_t}(t) \bar{V}(i_0, i_t). \quad (\texttt{gvar_i_residual}) \quad (4)$$

Define the standard centered prior covariance $\text{SmD} := \boldsymbol{\Sigma}_0 - \text{diag}(\boldsymbol{\mu}_0)$ (“SmD” in code). Instrument noise variance over the interval: $\epsilon_{0 \rightarrow t}^2 := \epsilon^2/t + \nu^2$ (denote it just e).

2 Observation model and the two scores

The macroscopic observation $\bar{y}^{\text{obs}}$ is modelled as heteroscedastic Gaussian:

$$\bar{y}^{\text{obs}} \mid p \sim \mathcal{N}(\mu_y(p), V(p)), \quad \mu_y(p) = N_{\text{ch}}(p \cdot \bar{\gamma}_0), \quad V(p) = e + N_{\text{ch}}(p \cdot \bar{\sigma}^2_0). \quad (5)$$

Both the mean and variance depend on the latent occupancy p . Centered innovation:

$$\delta(p) := \bar{y}^{\text{obs}} - \mu_y(p). \quad (6)$$

Two scores. The Gaussian log-likelihood factors into two orthogonal score components:

$$\boxed{\text{score}_{\mu}(p) := \frac{\delta(p)}{V(p)}} \quad (\text{score for the mean direction; Aitkin's } \eta), \quad (7)$$

$$\boxed{\text{score}_V(p) := \frac{\delta(p)^2}{V(p)} - 1} \quad (\text{score for the variance direction; Aitkin's } \zeta). \quad (8)$$

Under the model, $\mathbb{E}[\text{score}_\mu] = 0$, $\mathbb{E}[\text{score}_V] = 0$, $\mathbb{E}[\text{score}_\mu^2] = 1/V$, $\mathbb{E}[\text{score}_V^2] = 2$, $\mathbb{E}[\text{score}_\mu \text{score}_V] = 0$. The gradient of the log-likelihood with respect to p is

$$\nabla_p \log L = \text{score}_\mu \cdot \partial_p \mu_y + \frac{\text{score}_V}{2V} \partial_p V = N_{\text{ch}} \left[\text{score}_\mu \bar{\gamma}_0^T + \frac{\text{score}_V}{2V} \bar{\sigma}_0^T \right]. \quad (9)$$

This decomposition is the canonical entry point for any rank-2 Laplace update: score_μ drives the mean, score_V drives the variance, and they contribute through orthogonal directions in p -space.

3 The tilde operator: K -dim closed forms for K^2 -dim boundary objects

The full state-pair (boundary) covariance of channel counts is $K^2 \times K^2$ -dimensional. MacroIR/IRT never builds it: every quantity that the update needs is a closed-form contraction expressible in terms of μ_0 (K -dim), Σ_0 ($K \times K$), $\mathbf{P}$ ($K \times K$), and the boundary moment matrices $\bar{\Gamma}, \bar{V}$ ($K \times K$). The *tilde operator* packages these contractions.

For a boundary matrix $\bar{\mathbf{U}} \in \mathbb{R}^{K \times K}$ with per- i_0 marginal $\bar{\mathbf{u}}_0(i_0) = \sum_{i_t} P_{i_0 \rightarrow i_t} \bar{U}_{i_0 \rightarrow i_t}$, and the analog $(\bar{\mathbf{W}}, \bar{\mathbf{w}}_0)$:

Tilde scalar (closed form for $\widetilde{u^T \Sigma w}$).

$$\widetilde{u^T \Sigma w} := \bar{\mathbf{u}}_0^T \text{SmD} \bar{\mathbf{w}}_0 + \mu_0^T [\mathbf{P} \circ \bar{\mathbf{U}} \circ \bar{\mathbf{W}}] \mathbf{1}. \quad (10)$$

The first term is the per- i_0 ensemble-state contribution; the second is the boundary-state cross-correlation, computed as a triple elementwise product followed by a row sum. Both terms are $O(K^2)$.

Tilde vector (cross-covariance vector to the endpoint state). In the *endpoint frame* (matching the live kernel's gS, which is multiplied by $\mathbf{P}$ on the right so it lives at the post-Markov frame for the rank-1 down-date):

$$\widetilde{u^T \Sigma} := \bar{\mathbf{u}}_0^T \text{SmD} \mathbf{P} + \mu_0 (\mathbf{P} \circ \bar{\mathbf{U}}). \quad (11)$$

A $1 \times K$ row vector. $O(K^2)$.

Specializations needed for IRT.

$$\widetilde{\gamma \Sigma \gamma} = \widetilde{\gamma^T \Sigma \gamma} \quad (\text{use } \bar{\mathbf{U}} = \bar{\mathbf{W}} = \bar{\Gamma}), \quad (12)$$

$$\widetilde{\sigma \Sigma \sigma} = \widetilde{\sigma^T \Sigma \sigma} \quad (\text{use } \bar{\mathbf{U}} = \bar{\mathbf{W}} = \bar{V}), \quad (13)$$

$$\widetilde{\gamma \Sigma \sigma} = \widetilde{\gamma^T \Sigma \sigma} \quad (\text{use } \bar{\mathbf{U}} = \bar{\Gamma}, \bar{\mathbf{W}} = \bar{V}), \quad (14)$$

$$\widetilde{\gamma \Sigma} = \widetilde{\gamma^T \Sigma} \quad (\text{vector form, } \bar{\mathbf{U}} = \bar{\Gamma}), \quad (15)$$

$$\widetilde{\sigma \Sigma} = \widetilde{\sigma^T \Sigma} \quad (\text{vector form, } \bar{\mathbf{U}} = \bar{V}). \quad (16)$$

Why this is the value-add. Without the tilde operator, computing $\widetilde{u^T \Sigma w}$ would require the full $K^2 \times K^2$ joint state-pair covariance. Equation (??) expresses the same quantity using only μ_0 , SmD, and elementwise products of $K \times K$ matrices. The Woodbury inversion in Section ?? therefore factors through the $K \times K$ prior Σ_p and a 2×2 inner matrix — never a K^2 -dim object.

4 Predictive moments

$$\mu_y(p) = N_{\text{ch}} (p \cdot \bar{\gamma}_0), \quad (17)$$

$$V(p) = e + N_{\text{ch}} \widetilde{\gamma \Sigma \gamma}(p) + N_{\text{ch}} (p \cdot \overline{\sigma^2}_0). \quad (18)$$

The three terms in $V(p)$ are: instrument noise; boundary cross-correlation (via the tilde scalar); and the per- i_0 residual conductance variance contribution. At $p = \mu_0$ we obtain the predictive mean and variance:

$$\bar{y}^{\text{pred}} = N_{\text{ch}} \mu_0 \cdot \bar{\gamma}_0, \quad (19)$$

$$V = e + N_{\text{ch}} \widetilde{\gamma \Sigma \gamma} + N_{\text{ch}} \mu_0 \cdot \overline{\sigma^2}_0, \quad (20)$$

and the innovation $\delta = \bar{y}^{\text{obs}} - \bar{y}^{\text{pred}}$, the two scores from (??)–(??).

The IR over-count question, settled. Note that $\overline{\sigma^2}_0$ is the *residual* per- i_0 variance ($\sum_{i_t} P_{i_0 \rightarrow i_t} \bar{V}(i_0, i_t)$); the *variance-of-conditional-mean across end states* is already inside $\widetilde{\gamma \Sigma \gamma}$ via the elementwise-product term in (??). Using $\overline{\sigma^2}_0$ here (not $\text{Var}(|X_0=i)$) *is what the boundary derivation requires; the historical bug fixed at leg*

5 Energy, gradient, Hessian

We work with the energy $E(p) := -2 \log \text{posterior}(p) + \text{const}$:

$$E(p) = \log V(p) + \frac{\delta(p)^2}{V(p)} + (p - \mu^{\text{prior}}) [N_{\text{ch}} \Sigma_p^{-1}] (p - \mu^{\text{prior}})^T. \quad (21)$$

With $\nabla_p \delta = -N_{\text{ch}} \bar{\gamma}_0^T$ and $\nabla_p V = N_{\text{ch}} \overline{\sigma^2}_0^T$, the gradient of E is

$$\mathbf{g}(p) = 2N_{\text{ch}} (p - \mu^{\text{prior}}) \Sigma_p^{-1} - N_{\text{ch}} \left[2 \text{score}_\mu \bar{\gamma}_0^T + \text{score}_V \cdot V^{-1} \overline{\sigma^2}_0^T \right]. \quad (22)$$

At $p = \mu^{\text{prior}}$ the prior term vanishes:

$$\mathbf{g}_0 = -N_{\text{ch}} \left[2 \text{score}_\mu \bar{\gamma}_0^T + \text{score}_V \cdot V^{-1} \overline{\sigma^2}_0^T \right]. \quad (23)$$

Both scores enter explicitly. *This is the rank-2 origin*: score_μ and score_V drive separate directions, $\bar{\gamma}_0^T$ and $\overline{\sigma^2}_0^T$ respectively.

The Hessian of E (computed in detail in the previous archive supplement) has the structure

$$\mathbf{H}(p) = 2N_{\text{ch}} \Sigma_p^{-1} + \frac{2N_{\text{ch}}^2}{V} \mathbf{v} \mathbf{v}^T - \frac{N_{\text{ch}}^2}{V^2} \overline{\sigma^2}_0 \overline{\sigma^2}_0^T, \quad (24)$$

where $\mathbf{v} := \bar{\gamma}_0 + (\delta/V) \overline{\sigma^2}_0$ is the *tilted direction* that absorbs the cross-couplings between score_μ and score_V . The two outer products are exactly the rank-2 Hessian update of the prior precision: $\mathbf{v} \mathbf{v}^T$ is the mean-direction term, and $\overline{\sigma^2}_0 \overline{\sigma^2}_0^T$ is the variance-direction term. The sign of the second is negative because of the $\log V$ curvature.

6 Rank-2 Hessian as a Woodbury update

Set $\mathbf{U} := [\mathbf{v}, \overline{\boldsymbol{\sigma}^2_0}] \in \mathbb{R}^{K \times 2}$ and

$$\mathbf{C} := \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}, \quad \alpha := \frac{N_{\text{ch}}}{V}, \quad \beta := -\frac{N_{\text{ch}}}{2V^2}, \quad (25)$$

so that (??) becomes $\mathbf{H} = 2N_{\text{ch}}[\boldsymbol{\Sigma}_p^{-1} + \mathbf{U}\mathbf{C}\mathbf{U}^T]$.

Woodbury inversion gives

$$\mathbf{H}^{-1} = \frac{1}{2N_{\text{ch}}} \left[\boldsymbol{\Sigma}_p - \boldsymbol{\Sigma}_p \mathbf{U} \mathbf{K} \mathbf{U}^T \boldsymbol{\Sigma}_p \right], \quad \mathbf{K} := (\mathbf{C}^{-1} + \mathbf{U}^T \boldsymbol{\Sigma}_p \mathbf{U})^{-1} \in \mathbb{R}^{2 \times 2}. \quad (26)$$

The 2×2 inner matrix in tilde-scalar terms. The $\mathbf{U}^T \boldsymbol{\Sigma}_p \mathbf{U}$ block is exactly the 2×2 of pairwise tilde scalars between $\mathbf{v}$ and $\overline{\boldsymbol{\sigma}^2_0}$:

$$\mathbf{U}^T \boldsymbol{\Sigma}_p \mathbf{U} = \begin{pmatrix} \mathbf{v}^T \boldsymbol{\Sigma}_p \mathbf{v} & \mathbf{v}^T \boldsymbol{\Sigma}_p \overline{\boldsymbol{\sigma}^2_0} \\ \overline{\boldsymbol{\sigma}^2_0}^T \boldsymbol{\Sigma}_p \mathbf{v} & \overline{\boldsymbol{\sigma}^2_0}^T \boldsymbol{\Sigma}_p \overline{\boldsymbol{\sigma}^2_0} \end{pmatrix} = \begin{pmatrix} \widetilde{v^T \Sigma v} & \widetilde{v^T \Sigma \sigma} \\ \widetilde{\sigma^T \Sigma v} & \widetilde{\sigma^T \Sigma \sigma} \end{pmatrix}, \quad (27)$$

where the boundary lift expands $\mathbf{v}$'s tilde forms in terms of $\overline{\boldsymbol{\Gamma}}$ and $\overline{\mathbf{V}}$ via Section 3:

$$\widetilde{v^T \Sigma v} = \gamma \widetilde{\Sigma} \gamma + 2(\delta/V) \gamma \widetilde{\Sigma} \sigma + (\delta/V)^2 \sigma \widetilde{\Sigma} \sigma, \quad (28)$$

$$\widetilde{\sigma^T \Sigma \sigma} = \gamma \widetilde{\Sigma} \sigma + (\delta/V) \sigma \widetilde{\Sigma} \sigma. \quad (29)$$

All quantities entering $\mathbf{K}$ are $O(K^2)$ tilde scalars. The 2×2 inversion is analytic.

Naming the Laplace-corrected variance. Define

$$\boxed{V_{\text{lap}} := V + N_{\text{ch}} \widetilde{v^T \Sigma v}} \quad (\text{"}V_{\text{lap}}\text{"}; \text{replaces submission's } \mathbf{e_mu}). \quad (30)$$

This is the predictive variance evaluated at the Laplace point in the mean direction, in the rank-1 (mean-only) sense. The submission's $\mathbf{e_mu} = \mathbf{e} + N_{\text{ms}} \mathbf{0}$ arises from solving for a related positivity-preserving root and is asymptotically the same.

Naming the two Kalman-gain factors.

$$\boxed{K_{\text{mean}} := \frac{N_{\text{ch}}}{V_{\text{lap}}}} \quad (\text{mean-direction Kalman gain factor}), \quad (31)$$

$$\boxed{K_{\text{var}} := \frac{N_{\text{ch}}}{2V_{\text{lap}}^2 + N_{\text{ch}} \widetilde{\sigma^T \Sigma \sigma}}} \quad (\text{variance-direction; replaces } \mathbf{zeta}). \quad (32)$$

7 Posterior covariance and mean (rank-2 updates)

Posterior covariance. $\boldsymbol{\Sigma}^{\text{post}} = 2\mathbf{H}^{-1}$ becomes a *rank-2 down-date* of $\boldsymbol{\Sigma}_p$:

$$\boxed{\boldsymbol{\Sigma}^{\text{post}} = \frac{1}{N_{\text{ch}}} \left[\boldsymbol{\Sigma}_p - \boldsymbol{\Sigma}_p \mathbf{U} \mathbf{K} \mathbf{U}^T \boldsymbol{\Sigma}_p \right]}. \quad (33)$$

In tilde-vector form (each column of $\Sigma_p \mathbf{U}$ is a tilde vector):

$$\Sigma_p \mathbf{U} = [\widetilde{v^T \Sigma}, \widetilde{\sigma \Sigma}^T], \quad (34)$$

so the down-date involves only the tilde vectors $\widetilde{v^T \Sigma}$ and $\widetilde{\sigma \Sigma}$ (each $1 \times K$), and the 2×2 $\mathbf{K}$. *No K^2 -dim object is materialised.*

Posterior mean (Newton step from $p_0 = \mu^{\text{prior}}$). Substituting (??) into $\mu^{\text{post}} = \mu^{\text{prior}} - \frac{1}{2} \mathbf{g}_0 \Sigma^{\text{post}}$ and simplifying:

$$\mu^{\text{post}} = \mu^{\text{prior}} + \left[\text{score}_\mu \widetilde{v^T \Sigma}^T + \left(\text{score}_\mu K_{\text{var}} \widetilde{v^T \Sigma \sigma} + \frac{\text{score}_V}{2 V_{\text{lap}}} \right) \widetilde{\sigma \Sigma}^T \right] \Sigma_p \Sigma^{\text{post}}. \quad (35)$$

The structure is transparent in two-delta language:

- the $\text{score}_\mu \widetilde{v^T \Sigma}^T$ term is the *mean-direction Kalman update* (the rank-1 limit when $\overline{\sigma^2}_0 = 0$);
- the $(\text{score}_\mu K_{\text{var}} \widetilde{v^T \Sigma \sigma} + \frac{\text{score}_V}{2 V_{\text{lap}}}) \widetilde{\sigma \Sigma}^T$ term is the *variance-direction correction* arising from the $\log V(p)$ curvature; it has two parts:
 - $\text{score}_\mu K_{\text{var}} \widetilde{v^T \Sigma \sigma}$: a score_μ -driven shift along $\overline{\sigma^2}_0$ due to the cross-coupling in the rank-2 Hessian;
 - $\text{score}_V / (2 V_{\text{lap}})$: a score_V -driven shift along $\overline{\sigma^2}_0$ that is non-zero *even when $\delta = 0$* (small- δ asymmetry: large $|\text{score}_V|$ means the observation is more or less variable than expected, pulling the posterior toward states with smaller or larger predicted variance).

8 Reductions

The diamond reductions follow algebraically from the formulas above:

Drop $\overline{\sigma^2}_0 \rightarrow 0$ (no state-dependent noise) recovers MacroIR. Then $\mathbf{v} = \overline{\gamma}_0$, $\widetilde{\sigma \Sigma \sigma} = \widetilde{\gamma \Sigma \sigma} = 0$, score_V does not enter the update direction, $V_{\text{lap}} = V$, K_{var} becomes irrelevant. The posterior reduces to the standard MacroIR rank-1 Kalman update:

$$\Sigma^{\text{post}} = \frac{1}{N_{\text{ch}}} \Sigma_p - \frac{1}{V} \widetilde{\gamma \Sigma}^T \widetilde{\gamma \Sigma}, \quad (36)$$

$$\mu^{\text{post}} = \mu^{\text{prior}} + \text{score}_\mu \widetilde{\gamma \Sigma}^T \Sigma_p \Sigma^{\text{post}}. \quad (37)$$

Drop the boundary lift recovers MacroMRT. Replace each tilde scalar/vector with its per- i_0 ensemble counterpart: $\widetilde{\gamma \Sigma \gamma} \rightarrow \overline{\gamma}_0^T \text{SmD} \overline{\gamma}_0$, $\widetilde{\gamma \Sigma} \rightarrow \overline{\gamma}_0^T \text{SmD} \mathbf{P}$ (per- i_0 lifted to endpoint), $\overline{\sigma^2}_0$ total replaces $\overline{\sigma^2}_0$ residual for the M family (boundary cross-cov term absent from $\widetilde{\gamma \Sigma \gamma} \Rightarrow$ total variance is what's needed). The rank-2 algebra is identical; the cost is just the absence of the elementwise-product cross-term in (??).

Combined: $\overline{\sigma^2}_0 \rightarrow 0$ and drop boundary lift recovers MacroMR. Standard Moffatt 2007 update.

9 Implementation map (math $\leftrightarrow$ code)

Math	Code (proposed)	Existing in code
δ	<code>dy</code>	yes
V	<code>r_y_var</code>	yes
$\text{score}_\mu = \delta/V$	<code>score_mean</code>	no (compute on the fly)
$\text{score}_V = \delta^2/V - 1$	<code>score_var</code>	no
$\bar{\gamma}_0$	<code>t_gmean_i</code>	yes
$\bar{\sigma}^2_0$ (residual)	<code>gvar_i_residual</code>	added at <code>qmodel.h:4063</code>
$\widetilde{\gamma\Sigma\gamma}$	<code>tilde_gamma_Sigma_gamma</code>	yes (in <code>gSg</code> for <code>av=2</code>)
$\widetilde{\sigma\Sigma\sigma}$	<code>tilde_sigma_Sigma_sigma</code>	no (= old <code>sSs</code>)
$\widetilde{\gamma\Sigma\sigma}$	<code>tilde_gamma_Sigma_sigma</code>	no (= old <code>sSg</code>)
$\widetilde{\gamma\Sigma}$	<code>tilde_gamma_Sigma_vec</code>	yes (= existing <code>gS</code>)
$\widetilde{\sigma\Sigma}$	<code>tilde_sigma_Sigma_vec</code>	no (= old <code>sS</code>)
$\widetilde{v^T\Sigma v}$	<code>tilde_v_Sigma_v</code>	no
$\widetilde{v^T\Sigma\sigma}$	<code>tilde_v_Sigma_sigma</code>	no
$\widetilde{v^T\Sigma}$	<code>tilde_v_Sigma_vec</code>	no
V_{lap}	<code>V_lap</code>	no (= old <code>e_mu</code>)
K_{mean}	<code>K_mean</code>	no (= existing N/V factor)
K_{var}	<code>K_var</code>	no (= old <code>zeta</code>)
$\mathbf{K}$	<code>K_inner</code> (2×2)	no

Cost summary. Per interval, in addition to the existing MacroIR computations: one extra elementwise product over $\bar{\mathbf{V}}$ for each of $\widetilde{\sigma\Sigma\sigma}$, $\widetilde{\gamma\Sigma\sigma}$ (both $O(K^2)$); one tilde vector $\widetilde{\sigma\Sigma}$ ($O(K^2)$); one 2×2 matrix inverse; $O(K)$ scalar arithmetic. Total overhead is a small constant times the existing IR cost. No $K^2 \times K^2$ matrix is built or stored at any point.

10 Numerical and trust-region notes

V_{lap} positivity. V_{lap} is built from V (positive) plus a positive contribution $N_{\text{ch}} \widetilde{v^T\Sigma v}$. It stays positive provided the tilde scalar is non-negative, which it is in exact arithmetic. Numerical floor: clamp at a small positive value if cancellation kicks in.

$\mathbf{K}$ inner-matrix conditioning. The 2×2 matrix $\mathbf{K}^{-1} = \mathbf{C}^{-1} + \mathbf{U}^T \Sigma_p \mathbf{U}$ has a negative-on-diagonal element ($\beta^{-1} = -2V^2/N_{\text{ch}}$) and can become singular when the $\bar{\sigma}^2_0$ -direction's contribution exactly balances the prior. Submission code falls back to `vc=false` when this occurs (the `delta_emu ≤ 0` check); we should preserve that fallback.

Trust-region simplex shrink ($\alpha_\mu, \alpha_\sigma$). The existing MacroIR trust-region machinery (`macro_ir_variance_inf`) should be applied to the new mean update (??), treating the bracketed direction as the effective `chi_gS` of the standard rank-1 update. The PSD canary α_σ similarly applies to the rank-2 down-date (??); concretely, the largest $\alpha \in [0, 1]$ keeping each diagonal of $\Sigma^{\text{prop}} - \alpha \cdot [\Sigma_p \mathbf{U} \mathbf{K} \mathbf{U}^T \Sigma_p]$ non-negative. Both are scalar minimisations over K elements; cost $O(K)$.

Single Newton step vs iterated MAP. Equation (??) is one Newton step from $\boldsymbol{\mu}^{\text{prior}}$. For typical N_{ch} , one step suffices. Iteration to convergence (the full MAP) would re-evaluate δ , V , $\mathbf{v}$, $\widetilde{\sigma\Sigma\sigma}$, etc. at each iterate and apply (??)–(??) with the updated quantities. The recipe of the original MacroTaylor supplement applies unchanged.

11 Diamond at a glance

	$\overline{\sigma^2}_0$ in V ?	score $_V$ -score?	boundary lift?	rank
MacroMR	yes (total)	no	no	1
MacroMRT	yes (total)	yes	no	2
MacroIR	yes (residual)	no	yes	1
MacroIRT	yes (residual)	yes	yes	2

The two horizontal arrows (M→I) add the boundary-state lift via the tilde operator. The two vertical arrows (no-T→T) add the variance-direction score score $_V$ and the rank-2 Hessian via the log V term. Pairwise contrasts on the diamond decompose any IRT-vs-MR gain into the two contributions independently.